

Task 2 — breaking it on purpose

Anvar · Lab 1

Compiled each one with `g++ <file>.cpp -o <out>`. Same errors show up in VS Code's terminal.

Error 1 — forgot a semicolon

Deleted the `;` at the end of line 4.

```
break1_missing_semicolon.cpp: In function 'int main()':
break1_missing_semicolon.cpp:4:46: error: expected ';' before 'return'
   4 |         std::cout << "Hello, Anvar!" << std::endl
       |                                         ^
       |                                         ;
   5 |         return 0;
       |         ~~~~~
```

What happened: I forgot the semicolon at the end of that `cout` line. C++ needs one to know a statement is finished, and since I left it off, the compiler just kept reading straight into the next line. It only got confused once it hit `return` — so the error points there instead of at the line I actually messed up. Took me a second to realize the real problem was one line up.

Error 2 — used a variable I never made

Typed `name` in the cout line but never declared it anywhere.

```
break2_undeclared_variable.cpp: In function 'int main()':
break2_undeclared_variable.cpp:4:31: error: 'name' was not declared in this scope;
did you mean 'tzname'?
   4 |         std::cout << "Hello, " << name << "!" << std::endl;
       |                                   ^~~~
       |                                   tzname
```

What happened: I used `name` in the cout line without ever declaring it first. The compiler had no idea what I meant, so it took a guess — it suggested `tzname`, some unrelated system variable that just happens to look similar. Definitely not what I meant, but I got a laugh out of that one.

Error 3 — missing closing brace

Deleted the `}` that closes `main()`.

```
break3_mismatched_brace.cpp: In function 'int main()':
break3_mismatched_brace.cpp:5:14: error: expected '}' at end of input
    5 |         return 0;
      |                     ^
break3_mismatched_brace.cpp:3:12: note: to match this '{'
    3 | int main() {
      |           ^
```

What happened: I deleted the closing brace on `main()` by accident. Since every `{` needs a matching `}`, the compiler reached the end of the file still thinking it was inside `main()` — so instead of blaming one specific line, it pointed all the way back to the opening brace I never closed.